VICTORY 2026

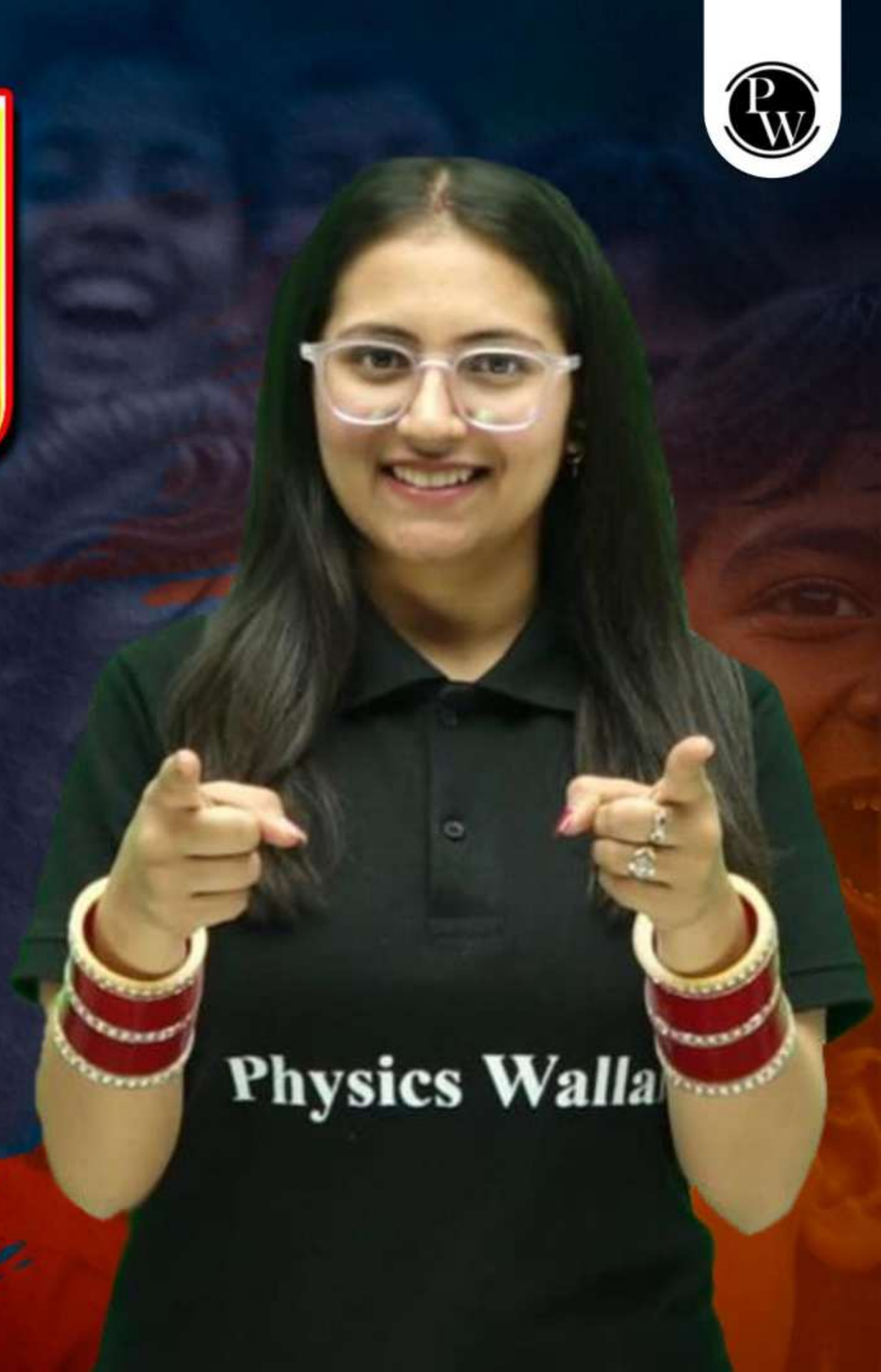
CHEMISTRY

Study of Compound:

NH_3 Ammonia

Lect.-02

By- Bharti Ma'am



Topics *to be covered*

1. Absorption of HCl gas in water $\longrightarrow$ HCl (acid)
2. Que: HCl
3. Basics: NH_3 (str., Properties)
4. Prep^{n}. of NH_3

Question

Name the glassware labelled X in the following figure.

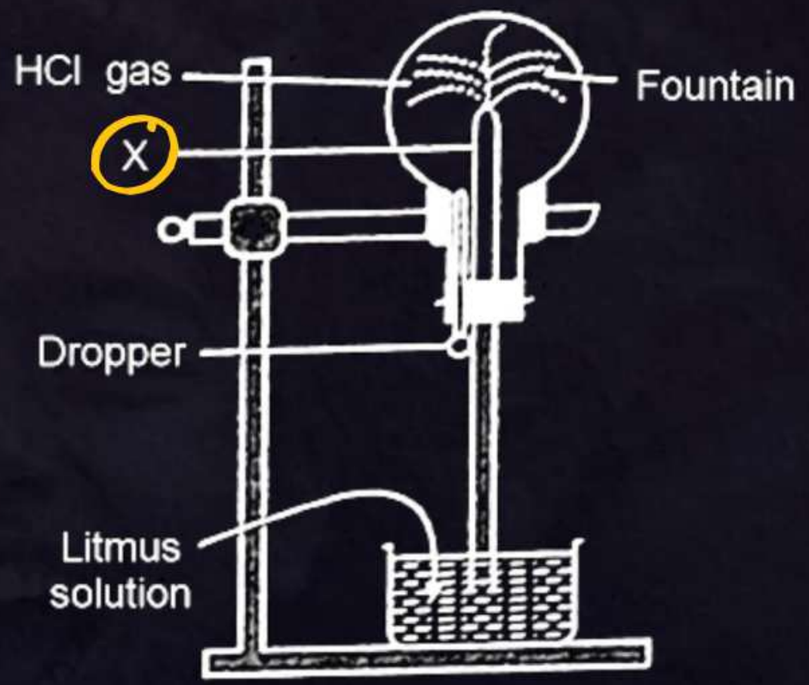


A Trough

B Jet tube

C Glass rod

D Plastic pipe



Question



Which of the following is/are the benefits of using funnel arrangement in the production of HCl acid

(i) It prevents back suction of water
(ii) If increases the temperature of a delivery, tube
(iii) It provides a large surface area

A (i) only

B (iii) Only

C (i) and (ii)

D (i) and (iii)

Question



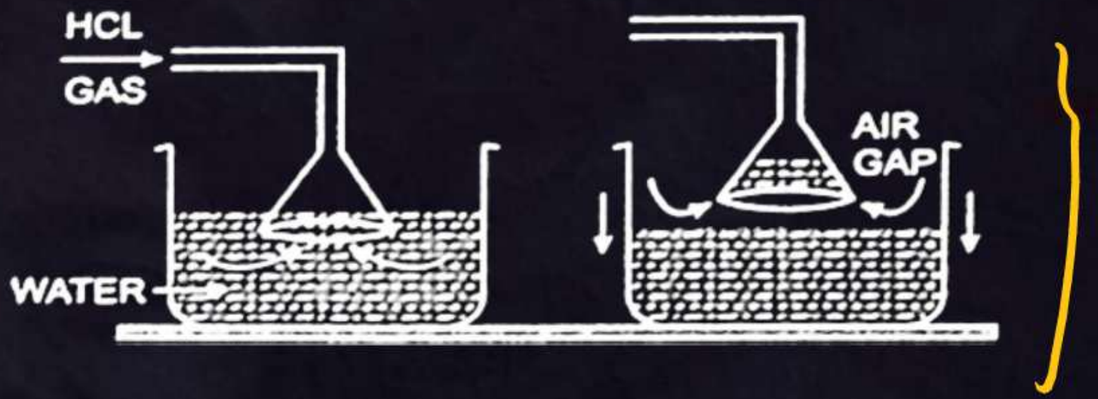
The funnel is used in the following experiment to ..... .

A Reduce temperature

B Decrease surface area

C Creates volume difference

D Avoid back suction



Question

Effect of hydrochloric acid on methyl orange is represented by



A Colour change from to orange red

B Colour change from red to orange

C Colour change from pink to colourless

D Colour change from colourless to pink

H.W

Hint:→ HCl (Acid)

Preparation of Hydrochloric Acid By Laboratory Method

- The aqueous solution of hydrogen chloride is known as hydrochloric acid. When hydrogen chloride gas bubbles through water it dissolves to form hydrochloric acid.

$$HCl(g) + H_2O \xrightarrow[\text{(dissolve)}]{} \text{Hydrochloric acid}$$

Preparation of Hydrochloric Acid By Laboratory Method

Method:

To convert gaseous HCl into aqueous form, dissolved it in water using an inverted funnel arrangement.

There are some steps involve in this method are as follows:

i) Place in inverted funnel over a container. Make sure that the rim of funnel points downward in the container.



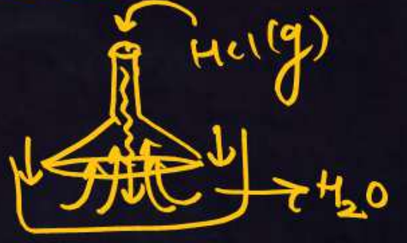
ii) Pour a sufficient amount of water in a container.

Preparation of Hydrochloric Acid By Laboratory Method

Method:

To convert gaseous HCl into aqueous form, dissolved it in water using an inverted funnel arrangement.

iii) Gently release the gaseous HCl near the opening of the inverted funnel. The gas will be carried downward by water.



iv) The HCl gas will dissolve in water, the water rises up in the funnel, back suction occurs în turn, the level outside the funnel falls, air gap between the rim of funnel and water level arises.

Preparation of Hydrochloric Acid By Laboratory Method

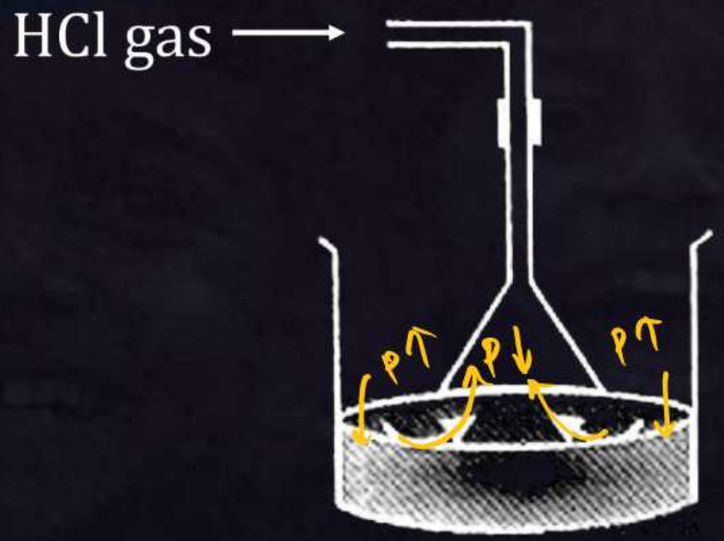
Method:

To convert gaseous HCl into aqueous form, dissolved it in water using an inverted funnel arrangement.}

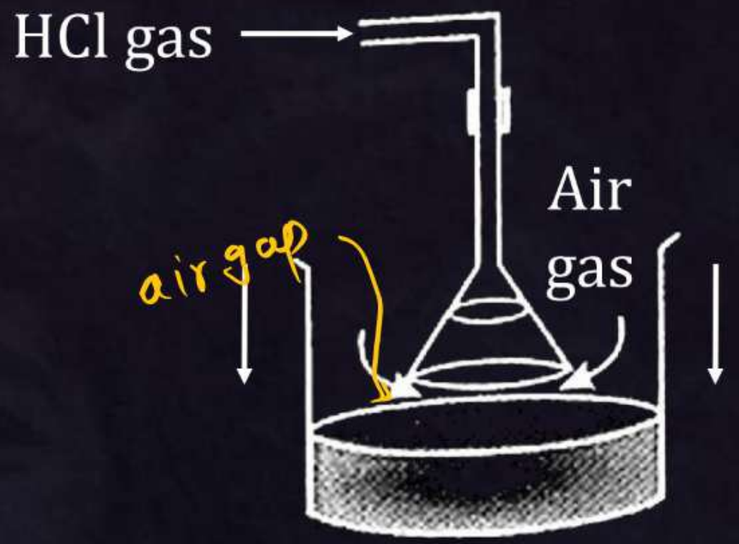
v) When pressure outside and inside become equal the water of funnel falls down again.

vi) This process continues in the beaker till saturated solution of HCl is not formed.

Tests for hydrogen chloride gas and Hydrochloric acid



(a) Water level in contact with the rim of the funnel



(b) Water level loses contact with the rim of the funnel

Fig. Funnel arrangement used for dissolving hydrogen chloride in water

The funnel management is done for the following. purposes:

1. **To minimise or prevent back section of water:** HCl gas dissolves in water more quickly than it is being generated in the flask. Due to this the pressure inside the funnel decreases and outside atmospheric pressure forces the water back up to the delivery tube. This effect is known as back suction)

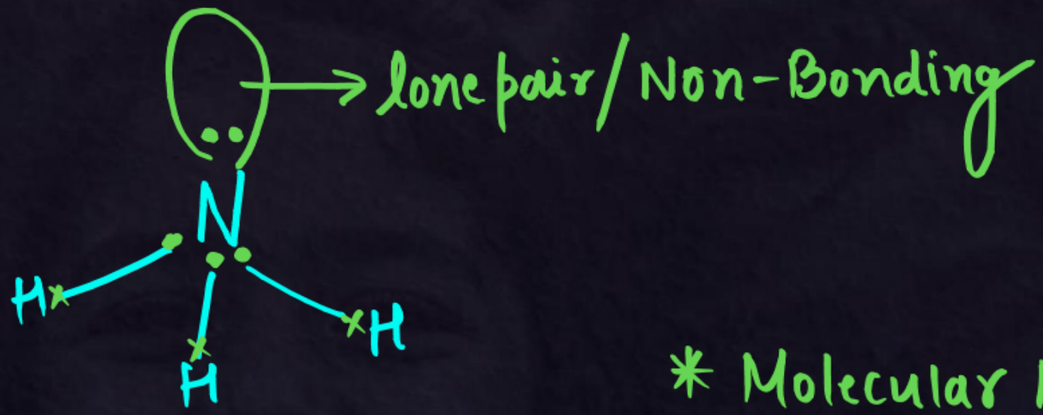
2. **To provide large surface area:** The shape of funnel increases the surface area for the intersection between the HCl gas and the liquid (usually water). This allows more efficient dissolution of the gas into the liquid.

3. **Minimize splashing:** The funnel helps minimize splashing and ensure a controlled entry of the gas into the liquid, reducing the risk of splits or accidents.

Ammonia

Chemical formula: NH_3

Structure:-



Covalent Bond

- Bond pair → 3
- Lone pair → one

* Molecular Mass NH_3

$= 14 + 3 \times 1$

$= 14 + 3$

$\Rightarrow$ 17 u/amu

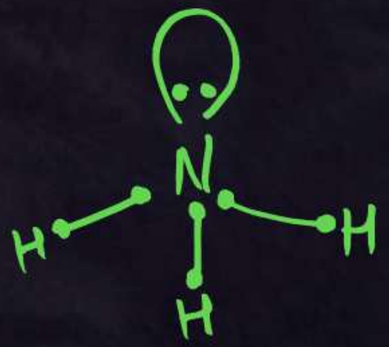
Que Total Bonding e^- in NH_3?

(A) $3e^-$

(B) $6e^-$

(C) $10e^-$

(d) Nhi Btayege



3 pair of e^-

Total e^-

1 pair = $2e^-$

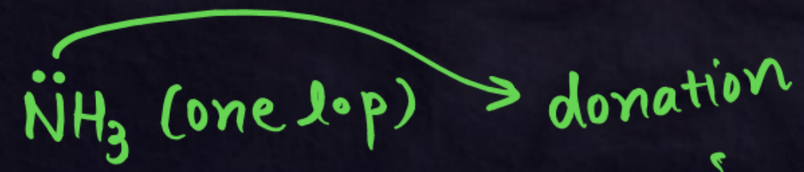
3 pair = 2×3

= $6e^-$

Lewis-Acid and Base Concept :-

1.) Acid OR Lewis acid = e^- Acceptor / gain

2.) Base OR Lewis Base ⇒ e^- donate / loss of e^-



NH_3 :→ Base

B.M vision



Forms of Ammonia

Ammonia is used commonly in the following forms:

(i) Gaseous ammonia (dry ammonia gas)

(ii) Liquid ammonia. When dry ammonia is liquefied by applying high pressure it is called liquid ammonia.



Forms of Ammonia

iii) Liquor ammonia fortis is a saturated solution of ammonia in water. It is called 880 ammonia. It has a relative density of 0.880. It is stored in tightly stoppered bottles in a cold place.

iv) Laboratory bench reagent: Dilute solution of liquor ammonia is used as a laboratory reagent.



General methods of preparation

- Ammonia gas can be prepared by warming an ammonium salt with caustic alkali, such as slaked lime or caustic soda or caustic potash.

Ammo. salt $\longrightarrow NH_3 \uparrow (g)$

$+$

Alkali

(KOH / NaOH)

General methods of preparation

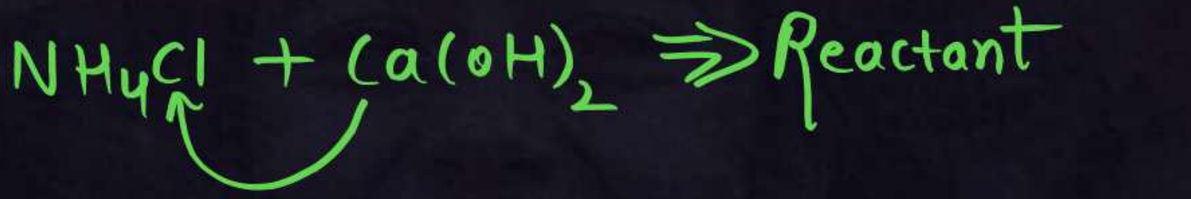
Ammonium salt	Alkali $\xrightarrow{\Delta}$	Salt +	Water +	Ammonia gas
$NH_4Cl(s)$ +	$Ca(OH)_2(s)$ Slaked lime $\xrightarrow{\Delta}$	$CaCl_2$	H_2O	$NH_3(g)\uparrow$
$(NH_4)_2SO_4$ +	$Ca(OH)_2$ $\xrightarrow{\Delta}$	$CaSO_4$	H_2O	$NH_3(g)\uparrow$
$(NH_4)_2SO_4$ +	$2KOH$ Caustic potash $\xrightarrow{\Delta}$	K_2SO_4	H_2O	$NH_3(g)\uparrow$

gen. Rxn

Laboratory preparation

1. **From ammonium chloride**

   Reactants: Ammonium chloride (NH_4Cl) and calcium hydroxide [$Ca(OH)_2$]. (Excess of calcium hydroxide is mixed well with ammonium chloride).

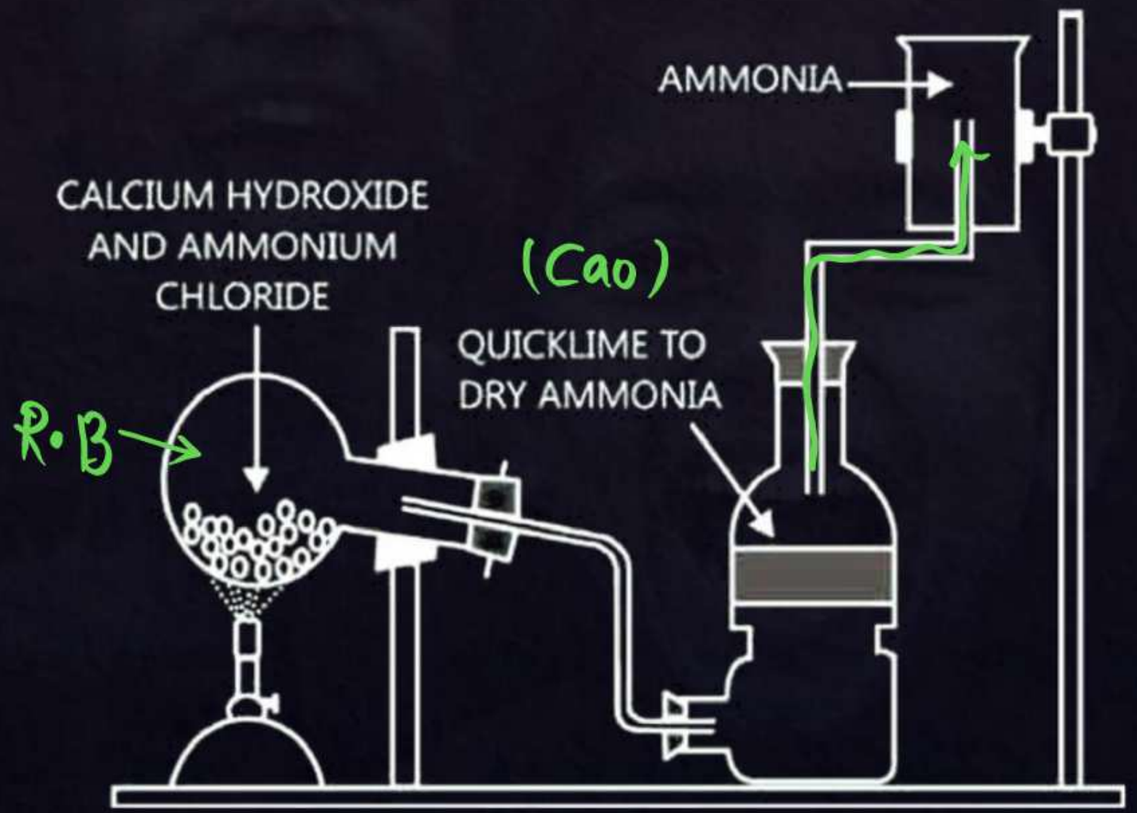


Laboratory preparation

Reaction:

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3 \uparrow$$

V. Imp.



Cao is a drying Agent

Laboratory preparation

Reaction:

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3 \uparrow$$

✓ Imp

(a) NH_4Cl

(b) $Ca(OH)_2$ (Alkali)

Laboratory preparation

Procedure:

Reactants (ammonium chloride and calcium hydroxide in excess) are finely ground and taken in a round-bottom flask fitted in a slanting position, mouth downwards. On heating the mixture, ammonia gas is evolved.

Laboratory preparation

V. Imp.

2. **Precautions:** The flask is tilted in such a way that the water formed in the reaction does not trickle back into the heated flask and thus break it.

Drying of ammonia gas: In order to get dry ammonia, the gas is passed through a drying tower containing lumps of quicklime (CaO).

Question



Can we use conc. H_2SO_4 as drying Agent in place of CaO

A Yes

B No

C May be

D Nhi pta

NH_3 (Basic) + H_2SO_4 (Acid) → Rxn



Laboratory preparation

V. Imp

Unsuitability of other drying agents: Other drying agents like conc. sulphuric acid, phosphorous pentoxide and anhydrous calcium chloride are not used, as ammonia being basic, reacts with them.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

$$6NH_3 + P_2O_5 + 3H_2O \rightarrow 2\ (NH_4)_3PO_4$$

ammonium phosphate

PYQ

$$CaCl_2 + 4NH_3 \rightarrow CaCl_2.4NH_3$$

(addition compound)



Laboratory preparation

- Ammonia gas is collected in inverted gas jars by the downward displacement of air because it is:

Imp

i) Lighter than air

ii) Highly soluble in water and therefore, it cannot be collected over water.



Collection

Note:

1. Higher ratio by weight of the alkali is used it may counteract the loss by sublimation of NH_4Cl.

2. Calcium hydroxide is used, as it is cheap and not deliquescent like other alkalies.

NH_4Cl → Sublimable compound

Sublimation :→ Solid → gas



Collection

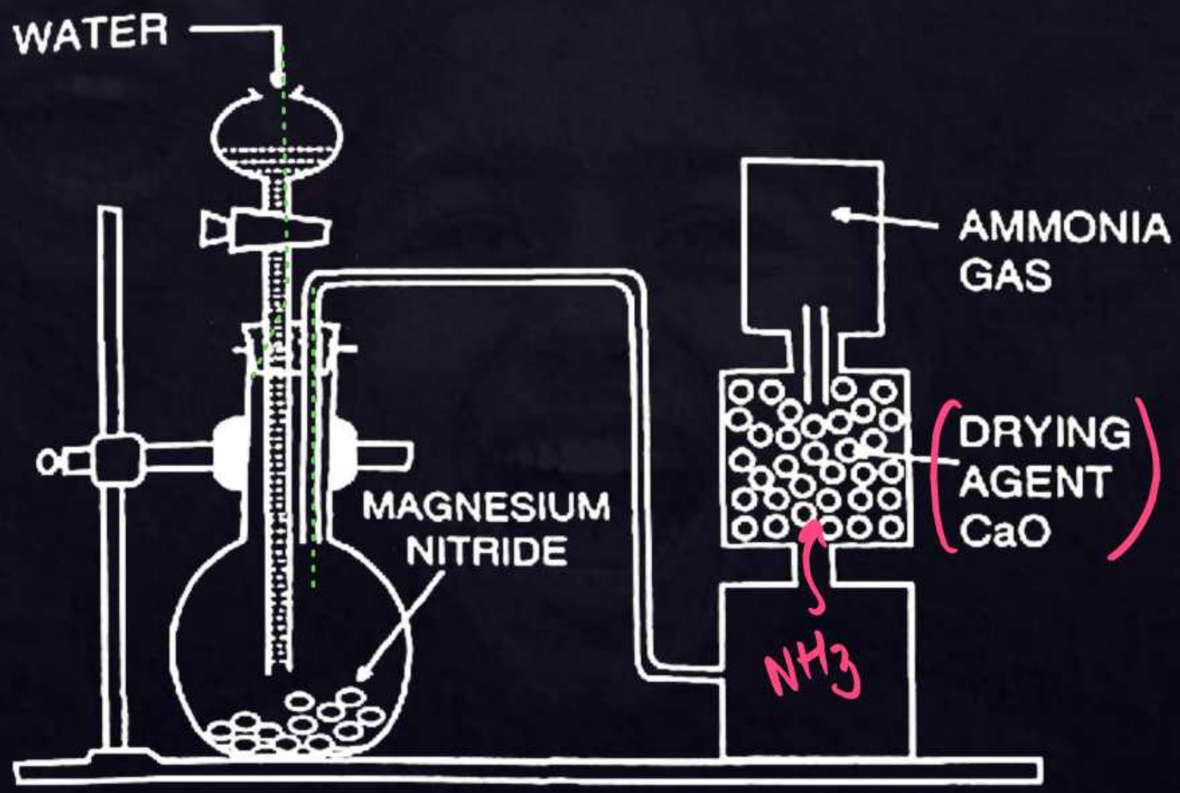
3. Though all ammonium salts, on heating with alkalies give NH_3, yet NH_4NO_3 (explosive in nature) and NH_4NO_2 on warming with alkalies do not produce ammonia because they decompose on heating.

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

Preparation of NH_3 by Metal Nitride

1. **Procedure:** Magnesium nitride or Aluminium nitride is kept in a round bottomed-flask fitted with a thistle, funnel and a delivery tube, as shown in Fig. Warm water is added very slowly.





Other MOP of NH_3

From metal nitrides

Ammonia can also be obtained by the action of warm water on nitrides of metals like magnesium or Aluminium. (This method is costly).

Reaction:

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3\uparrow$$

Or

$$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3\uparrow$$



Collection

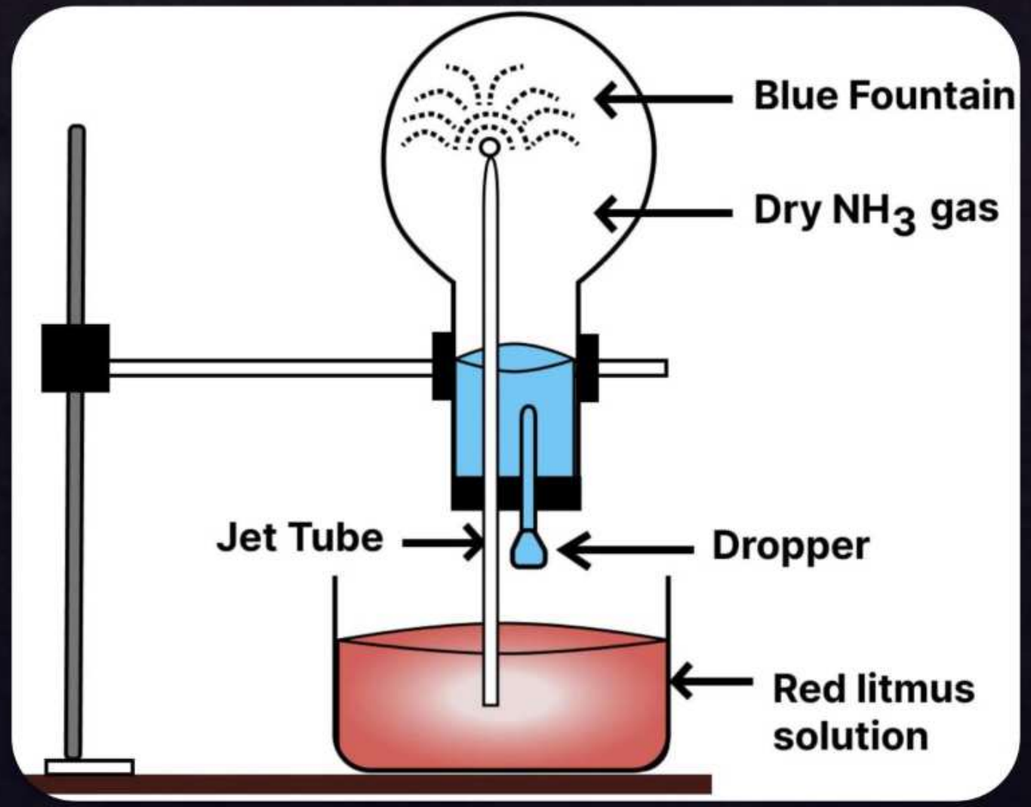
2. Ammonia gas is liberated and magnesium hydroxide or aluminium hydroxide is also formed, but it remains in the flask and the liberated ammonia gas escapes through the delivery tube.

3. The ammonia gas formed is collected by holding the jar with its mouth downwards over the delivery tube.

4. The gas can be dried by passing it through quicklime packed in a drying tower.

* Collection of Ammonia gas by downward displacement of Air

Fountain Experiment



Question



The property exhibited by concentrated sulphuric acid when it is used to prepare hydrogen chloride gas from potassium chloride:

[ICSE 2023]

A Dehydrating property

B Drying property

C Oxidizing property

D Non-volatile acid property

Hw

Question



The metallic oxide which when reacts with HCl forms salt and water:

[ICSE 2023]

A Carbon monoxide

B Nitrous oxide

C Ammonium hydroxide

D Sodium oxide

Question

Hydrogen chloride gas is not collected over water as: [ICSE 2022]



A. It is highly soluble in water

B. It is less soluble in water

C. It is lighter than air

D. It is heavier than air

Question



Hydrogen chloride can be obtained by adding concentrated sulphuric acid to:

A $NaCl$

B Na_2CO_3

C Na_2SO_4

D Alkali metals

Question

The activity of the Fountain Experiment is to prove that: [ICSE 2016]



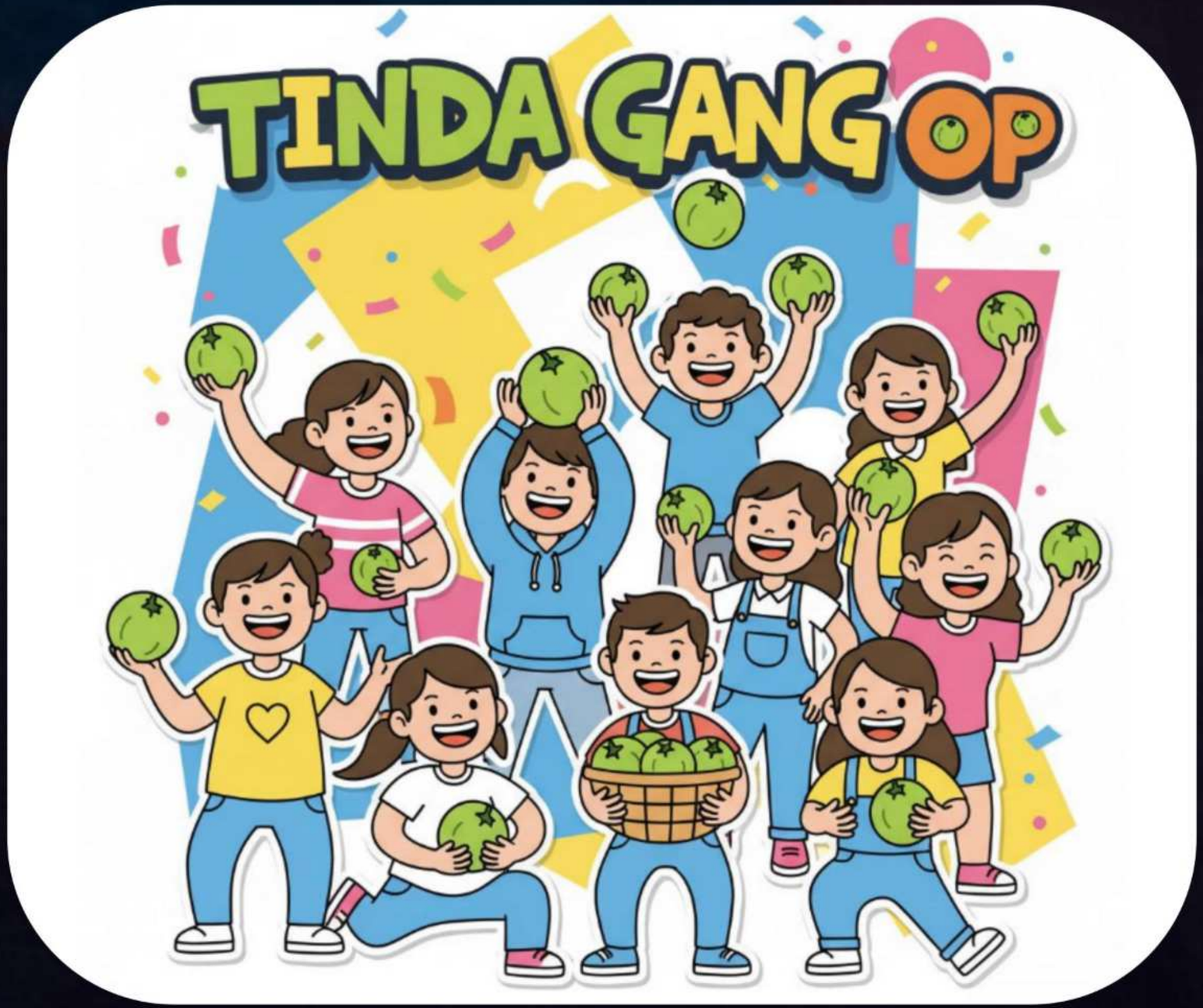
A HCl turns blue litmus red

B HCl is denser than air

C HCl is highly soluble in water

D HCl fumes in moist air


BHARTI MAAM
JOIN MY OFFICIAL TELEGRAM CHANNEL
Physics Wallah
@CHEMISTRYBYBHARTIMAAM


TINDA GANG OP


Thank You